INCH-POUND

MIL-H-24136A(SH)
AMENDMENT 1
21 September 1990

MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES, GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-H-24136A(SH), dated 18 June 1984, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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3.4.5: Delete and substitute:

"3.4.5 Concentricity. The maximum and minimum wall thickness shall be the maximum and minimum measurement obtained from a series of measurements taken around the circumference of the hose but concentrated in the area visually suspected of being thick or thin. The difference between the minimum and maximum wall thickness determined, is wall concentricity. The hose wall thickness shall be measured in accordance with ASTM D 380. Measurements shall be taken approximately 1/2 inch from hose end and measured to the nearest 0.001 inch. Concentricity shall not exceed the following values: For sizes 1/4-inch, 0.060 inch; for sizes over 1/4-inch up to and including 2 inch, 0.100 inch; for sizes over 2-inch, 0.125 inch."

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4.6.1: Delete and substitute:

"4.6.1 Proof test.

4.6.1.1 Proof pressure for less than 2-inch hose. The proof pressure test of the hose less than 2 inches in inside diameter shall be conducted in accordance with ASTM D 380 except that the samples shall be in accordance with 4.3.1 and 4.4.1 and the pressure shall be increased at a rate not less than 15,000 lb/in^2 per minute nor greater than 25,000 lb/in^2 per minute. The proof pressure shall be 200 percent of the maximum working pressure specified in the applicable specification sheet, and shall be held for not less than 1 minute nor more than 5 minutes. Hoses which pass during this test are acceptable for use. Evidence of leakage, rupture, or deformation of the hose shall constitute failure of this test.

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4.6.1.2 Proof pressure for greater than 2-inch hose. The proof pressure test of the hose over 2 inches inside diameter shall be conducted in accordance with ASTM D 380 except that the samples shall be in accordance with 4.3.1 and 4.4.1 and the pressure shall be increase at a uniform rate not to exceed 1000 lb/in^2 per minute. The proof pressure shall be 200 percent of the maximum working pressure specified in the applicable specification sheet, and shall be held for not less than 1 minute nor more than 5 minutes. Hoses which pass during this test are acceptable for use. Evidence of leakage, rupture, or deformation of the hose shall constitute failure of this test."

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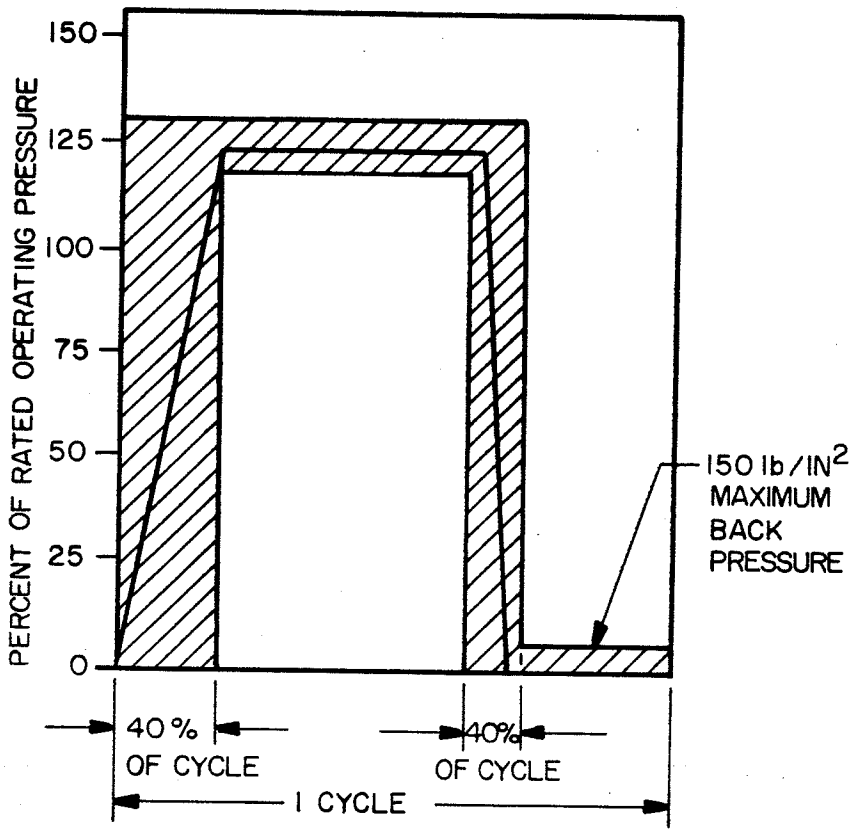
4.6.5: Delete and substitute:

"4.6.5 Impulse. Hose assemblies shall be subjected to impulse testing to the peak pressure specified on the applicable specification sheet. Hose assemblies shall be subjected to impulse rate from 30 to 80 cycles per minute, with the hydraulic fluid at a temperature of not less than 120°F nor greater than 130°F for a total of 150,000 cycles. Hose assemblies in sizes -4 through -12 shall be bent through 180 degrees to the minimum bend radius specified in the applicable specification sheet during the impulse test, with both ends rigidly supported. Hose assemblies larger than size -12 shall be dynamically impulsed straight. One end of the hose must be free to move during impulsing. Test fluid may flow through the test sample in order to maintain temperature as long as connections do not hamper free movement of one end of the hose. At the beginning of each pressure cycle, the peak pressure specified on the applicable specification sheet shall be attained. The applicable working (service) pressure shall be attained before leveling off (see figure 2, 3, or 4 as applicable)."

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FIGURE 4: Add new figure, attached:



For hose sizes larger than 5 inches (-80)

NOTE: The curve is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.6.5. The actual pressure time cycle obtained when performing this test, should fall within the shaded area.

FIGURE 4. Pressure time curve for impulse test.

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